B.Sc. Engg. CSE 5th Semester

11 April 2025

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

WINTER SEMESTER, 2023-2024
FULL MARKS: 150

CSE 4503: Microprocessor and Assembly Language

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all** 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

---

1. a) What was the alternative of Interrupt? Explain and classify different types of Interrupt. 3 + 6 (CO1) (PO1)

   b) What is Duty Cycle? Find out the relation between Clock State, Bus/Machine Cycle and Instruction Cycle. 3 + 6 (CO1) (PO1)

   c) Compare and contrast between the control signal operations of Maximum and Minimum modes of 8086 microprocessor. 7 (CO4) (PO2)

2. a) Draw the details of WRITE Bus timing diagram showing all the necessary signals of 8086. You should consider that the write operation is to be made to an output device with port address of 0FFFFh. Consider that there is 4 WAIT states, interrupt flag is SET and DS register is in use for data accessing. 9 (CO4) (PO2)

   b) What is MSW register? Explain the functionalities of IOPL, NT, and VM flags. 3 + 6 (CO1) (PO1)

   c) Suppose, while debugging an assembly language program the values of the registers are: Flag=ABCDh, IP=0102h, CS=8000h, SP=FFFAh. Now, if INT 21h is requested, derive the memory addresses from where the new IP and CS can be retrieved. Also, show the new SP value and steps involved in handling the interrupt by the 8086 microprocessor. 7 (CO4) (PO2)

3. a) How can multi-tasking be ensured with protected mode of a microprocessor? Differentiate between Real-mode, Protected Mode, and VM8086 Mode. 3 + 6 (CO2) (PO2)

   b) Draw the structures of memory banks of 80286, 80386, and Pentium. Mention how the address pins are used to identify the memory banks. 3 + 6 (CO3) (PO1)

   c) "A Pentium processor is 5 times faster than the 80486 numeric processor" – Justify the statement with an appropriate example. 7 (CO3) (PO1)

4. a) What is memory segment overlapping? Write short notes on logical address, linear address, and physical address. 3 + 6 (CO3) (PO1)

   b) What is Paging? Describe the process of calculation of linear address while paging is enabled and disabled. 3 + 6 (CO2) (PO1)

   c) Suppose, you are given a memory address of 4GB (with addresses 00000000h to FFFFFFFFh). Using a 16-bit segment register, clearly explain the process to define a global descriptor table along with local descriptor table. 7 (CO4) (PO2)

5. a) Drawing the architectures differentiate between CPU and GPU? 9 (CO3) (PO1)

b) "GPU is designed by optimizing few traditional CPU components" - What are those components and how are those optimized in GPU? 9 (CO3) (PO1)

c) Discuss the "Single/Multiple Instruction Stream and Single/Multiple Data Stream" considerations for designing single/multi-core processors. 7 (CO3) (PO3)

6. a) Write short notes on the followings using appropriate example/figure: 5×5 (CO1) (PO1)

   i. Thread
   ii. Hyper-thread
   iii. Turbo-boost
   iv. Scalar Processor
   v. CUDA Framework

B.Sc. Engg. CSE 5th Semester 06 December 2022

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2021-2022**
DURATION: 3 HOURS **FULL MARKS: 150**

CSE 4503: Microprocessor and Assembly Language

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer all 6 (six) questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

---

1. a) How the main memory of 80286, 80386 and Pentium processors are segmented? Which processor include the parity-bit at memory segment for error-control and how? 5+5 (CO3) (PO1)

   b) "80186 is more of a microcontroller than a microprocessor" – True/False? Distinguish between the DX and SX version of 80386 microprocessor. 2+8 (CO3) (PO1)

   c) Suppose, while debugging an assembly language program the values of the registers are: `Flag=FEB9h`, `IP=0102h`, `CS=0700h`, `SP=FFFAh`. Now, if `INT 00100001B` is requested, derive the memory addresses from where the new `IP` and `CS` can be retrieved; Also show the step-by-step changes in memory contents of stack segment along with corresponding `SP` values while handling the interrupt by the 8086 microprocessor. 5 (CO1) (PO1)

2. a) What do you mean by Coppermine? How do Coppermine and L2 cache memory differ from each other? 2+8 (CO3) (PO3)

   b) Briefly explain the operations of IOPL and NT flags of 80286 microprocessor. 5+5 (CO3) (PO1)

   c) Let, in an 8086 assembly language code's data segment is initialized at `0700h:0000h` memory location and variables are declared as follows: 5 (CO4) (PO1)

```
.DATA
A db 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
B db 'IUTOIC'
C dw ABCDh
D db 'Exam Month is November/December, 2022','$'
```

   Now, using appropriate instructions derive and store the initial offsets of variable A, B, C and D at SI, DI, BX, BP registers, respectively. Also, derive the DS register value.

3. a) "Parallel programming ensures higher utilization of multi-core processors" – How? Explain with an example. 10 (CO3) (PO1)

   b) What is Multi-core Processor? What performance issues actually force to design multi-core processor system? 2+8 (CO3) (PO1)

   c) Write one or more 8086 instruction(s) sets using appropriate register(s) to do the followings: 5 (CO4) (PO4)
      i. Multiply the value of 8192 by 8 and show the result.
      ii. Divide the value of 254 by 2 and show the quotient and remainder results.

4. a) What is Virtual 8086 Mode? Explain the operations of GDT and LDT registers for memory segmentation. 2+8 (CO3)

(PO1)

b) Draw the details of WRITE Bus timing diagram showing all the necessary/required signals of 8086. You should consider that the write operation will be made in an output device with port address of 01111H and there is no WAIT state. 10 (CO2) (PO1)

c) Write an assembly language program, where a MACRO is used to address a string and a PROCEDURE is used to display that string. 5 (CO4) (PO2)

5. a) Why do we need Interrupt for microprocessors? Differentiate between polling and interrupt. (4+6) 4+6 (CO2) (PO1)

b) What is Pipelining? Considering either U-pipeline or V-pipeline show the pipeline operation of Intel microprocessor. 2+8 (CO3) (PO2)

c) Define and Distinguish between the LEA and OFFSET instructions. 5 (CO4) (PO1)

6. a) Write short notes on the followings using appropriate example/figure: 5×5 (CO1) (PO1)
   i. Memory Segment Overlapping
   ii. Interrupt Vector Table
   iii. Signal Duty Cycle
   iv. Hyper-thread
   v. Turbo Mode

Name of the Program: B.Sc. in CSE/B.Sc. in TVE **16 September 2021**
Semester: Winter 2020-2021 **Time: 2:30 pm – 4:00 pm**

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

Semester Final Examination **Winter Semester: 2020-2021**

Course Number : CSE 4503/4573 **Full Marks: 75**
Course Title : Microprocessors and Assembly Language **Time: 1.5 Hours**

There are **3 (three)** questions. Answer all of them. Figures in the right margin indicate marks. The examination is **Online** and **Open Book**. Marks of each question and corresponding **CO** and **PO** are written in the brackets.

Write **Student ID** and **Name** top of the **first page** and write **student ID** and **page no** in every page of the answer script. Submission pdf of the answer script should be named as **Full_Student_ID<space>Course Code.pdf**

---

1. a) What are real mode and protected mode? Which microprocessor(s) does first implement the protected mode and how? 10 (CO4) (PO1)

   b) How does thread ensure faster processing? Discuss in the context of multi-core processor system? 10 (CO1) (PO1)

   c) 'Utilization of parallel processors can be achieved through parallel programming'. How? Prove with appropriate example. 5 (CO1) (PO1)

2. a) Which memory locations are reserved for 8086 Interrupt Vector Table? Derive the specific memory locations where the CS and IP values of the following Interrupt Types can be found: 10 (CO4) (PO2)
   i. `INT 10B`
   ii. `INT 3`

   b) Let, in an assembly language code data segment is initialized at `0700h:0000h` memory location of 8086 and variables are declared as follows: 10 (CO1) (PO1, PO2)

```
.DATA
A db 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
B db 'IUTOIC'
C dw 1234
D db 'Exam Date is 16 September, 2021','$'
```

   Now, using appropriate instructions derive and store the initial offsets of variable A, B, C and D at SI, DI, BX, BP registers, respectively. Also, derive the DS register value.

   c) Write an assembly language code to take a single-character (digits 0 to 9 only) as an *input* and display the previous character (as in ASCII table) as an *output* in the new line with carriage return. 5 (CO1) (PO1)

3. a) Draw the details of WRITE Bus timing diagram showing all the necessary/required signals of 8086. You should consider that the write operation will be made in the I/O port address of `1111h` and there are 3-T states used in WAIT state. 10 (CO4) (PO2)

   b) Let, in 8086 based system current SP value is `FFFAh` and the CS, IP and FL values are `7000h`, `1000h` and `1001h`, respectively. Now, if at that instance an interrupt occurs, then show the stack increase and decrease scenario along with changes in SP values and stack contents. 10 (CO2) (PO1, PO2)

c) Define the following terms with appropriate example: 5 (CO2) (PO2)

i. Clock cycle

ii. Bus cycle

iii. Instruction cycle

iv. Machine cycle

v. ‘T’ duration (consider a microprocessor is having a clock speed of 12 MHz)

B.Sc. Engg. / HD CSE 5th Semester **24 August 2020 (Morning)**

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
<u>Department of Computer Science and Engineering (CSE)</u>

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2019-2020**

DURATION: 1.5 Hours **FULL MARKS: 80**

CSE 4503: Microprocessors and Assembly Language

Programmable calculators are not allowed. Do not write anything on the question paper.

There are **<u>5 (five)</u>** questions. Answer any **<u>4 (four)</u>** of them.

Figures in the right margin indicate marks.

---

1. a) Distinguish between the followings: 10
   i. Polling and Interrupt.
   ii. Memory-mapped I/O and Isolated I/O.

   b) Let, in an assembly language code data segment is initialized at `7000h:0000h` memory location of 8086 and variables are declared as follows: 10

```
.DATA
A db 1, 2, 3, 4, 5
B db 'IUT'
C dw FFFFh
D db 'Exam Date is 24 August, 2020','$'
```

   Now, using appropriate instructions derive and store the initial offsets of A, B, C and D at SI, DI, BX, BP registers, respectively. Also, derive the DS register value.

2. a) You have to draw a detail WRITE bus timing diagram for 8086 microprocessor; where, consider that data is transmitted toward a memory location only. 10

   b) With an appropriate timing diagram clearly define the terms: *Clock cycle*, *Bus cycle* and *Instruction cycle*. Also, find out the T durations of the microprocessors having clock speeds of 40 MHz and 1 GHz, respectively. 10

3. a) Which memory locations are reserved for 8086 Interrupt Vector Table? Derive the specific memory locations where the CS and IP values of the following Interrupt Types can be found: 10
   i. `INT 21h`
   ii. `INT 3`

   b) Let, in 8086 based system current SP value is `FFF8h` and the CS, IP and FL values are `7000h`, `0100h` and `1101h`, respectively. Now, if at instance an interrupt occurs, then show the stack increase and decrease scenario along with changes in SP values and stack contents. 10

4. a) Derive the contents of the following `MOV` instructions using its coding template and also show how the contents of the instructions can be stored in memory: 10
   i. `MOV DL, 00000001B`
   ii. `MOV SS:[SI], BP`

   b) Write an assembly language program that will display "`Islamic University of Technology`" 10 (ten) times with new lines each starts at home position. 10

5. a) Differentiate between different `80x86` microprocessors. 10

   b) Write an assembly language program structure to clearly state the operational differentiation between `LABEL` and `LOOP`? 10

316

B.Sc. Engg. / HD CSE 5th Semester 18 May 2019

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2018-2019**
DURATION: 3 Hours **FULL MARKS: 150**

CSE 4503: Microprocessors and Assembly Language

Programmable calculators are not allowed. Do not write anything on the question paper.
There are **8 (eight)** questions. Answer any **6 (six)** of them.
Figures in the right margin indicate marks.

---

1. a) What is machine language? How can we get machine language from an assembly language? Explain with an example. 10
   b) Briefly explain about multiple interrupt concepts. 8
   c) What are basic differences between `LOOP` and `LEVEL` in assembly language programming? 7

2. a) Derive the contents of the following `MOV` instructions using its coding template and also show how the contents of the instructions can be stored in memory: 12
      i. `MOV DX, BX`
      ii. `MOV AAFFh[DI], AH`
      iii. `MOV AX, [1234h]`
   b) Write an assembly language program that will display "Microprocessors and Assembly Language" 10 (ten) times in different lines with line feed and carriage return. 6
   c) Considering following memory segments, offsets and instructions, write the sequence of PUSH/POP operations on stack segment mentioning different Stack Pointer (SP) values. Assume, initially the stack segment is empty. 7

Segment	Offset	Assembly Language
1000h	0100h	IN AL, 27h
1000h	0102h	MOV DL, AL
1000h	0104h	MOV AH, 1
1000h	0106h	**INT 21h**
1000h	0108h	ADD AL, DL

3. a) Draw the coding template of `IN` instruction. Explain the significance of using '`MOD`' and '`R/M`' in `MOV` coding template. 9
   b) Write the equivalent assembly language code structures using *conditional jump* and *loop* instructions to implement the *if-else*, *for* and *while* loop operations. 9
   c) Suppose, while debugging an assembly language program the values of the registers are: `Flag=FEB9h`, `IP=0102h`, `CS=0500h`, `SP=FFFCh`. Now, if `INT 21h` is requested, derive the memory addresses from where the new `IP` and `CS` can be retrieved; Also show the new `SP` value and steps involved in handling the interrupt by the 8086 microprocessor. 7

4. a) Write short differentiations between the following 8086 assembly language instructions: 9
      i. `ROL and SHL` ii. `LEA and OFFSET` iii. `NOT and NEG`
   b) Narrate the function of using `1`, `2` and `9` under `INT 21h` instruction. 8
   c) Distinguish between Memory-mapped I/O and Isolated I/O. 8

5. a) Draw the bus timing diagram for a microprocessor's operation while it performs a WRITE operation toward an OUTPUT unit. 10
   b) What are the basic differences between MIN and MAX mode of 8086 pin diagram? 6
   c) In how many ways can you define an array using assembly language programming? Give example code for each of them. 9

6. a) Draw a comparative table to differentiate between the features of 8086, 80186 and 80286 microprocessors. 10
   b) 'Utilization of parallel processors can be achieved through parallel programming'. How? Prove with appropriate example. 8
   c) Write the functionalities of IOPL and NT flags for 80286 microprocessor. 7

7. a) What do you mean by Coppermine? How do Coppermine and L2 cache memory differ from each other? 8
   b) How are the main memory of 80386 and Pentium processors segmented? Mention the use of address bus pins for both 80386 and Pentium microprocessors. 9
   c) Write an assembly language program, where a MACRO is used to address a string and a PROCEDURE is used to display that string. 8

8. a) Define Thread and Turbo Mode in the context of multi-core processor system? 10
   b) Differentiate between the features of core i3, i5 and i7 processors. 9
   c) Write short notes on: 6
      i. U-Pipeline
      ii. V-Pipeline
      iii. Floating Point Unit (FPU)

301

B.Sc. Engg. / HD CSE 5th Semester 01 June 2018

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2017-2018**
DURATION: 3 Hours **FULL MARKS: 150**

CSE 4503: Microprocessors and Assembly Language

Programmable calculators are not allowed. Do not write anything on the question paper.
There are 8 (eight) questions. Answer any 6 (six) of them.
Figures in the right margin indicate marks.

---

1. a) What do you mean by single and multi-core microprocessor systems? Briefly explain the importance of using assembly language in a microprocessor system. 10

   b) Derive the contents of the `Flag` (`CF, PF, ZF, OF`) register of `8086` microprocessor upon executing the following instructions: 8

      i. `CMP AL, FFh` ; Assume `AL` initially contains `FFh`.
      ii. `TEST AL, FFh` ; Assume `AL` initially contains `FFh`.

   c) Explain the purpose of `DUP` operator with an example. 7

2. a) Derive the contents of the following `MOV` instructions using its coding template and also show how the contents of the instructions can be stored in memory: 12

      i. `MOV AL, BL`
      ii. `MOV FFh[SI], BH`
      iii. `MOV DX, [ABCDh]`

   b) Write short differentiations between the following 8086 assembly language instructions: 8

      i. `ROR` and `SHR` ii. `LEA` and `OFFSET` iii. `NOT` and `NEG`

   c) Write an assembly language program structure to allocate exactly 64 Kbytes of memory for *data segment*, default memory bytes for *stack segment* and also consider that the size for *code segment* may exceed 64 Kbytes. 5

3. a) Draw the schematic architecture of 8088 microprocessor. Write short notes on *segment registers* of 8086 microprocessor. 9

   b) Write an assembly language program that takes N as a decimal digit (0 ~ 9) input and shows the summation of $1+2+ \ldots + N$ as output. 9

   c) Suppose, while debugging an assembly language program the values of the registers are: `Flag=FEB9h, IP=0102h, CS=0500h, SP=FFFCh`. Now, if `INT 21h` is requested, derive the memory addresses from where the new `IP` and `CS` can be retrieved; Also show the new `SP` value and steps involved in handling the interrupt by the 8086 microprocessor. 7

4. a) Drawing the timing diagram, briefly explain the `READ` and `WRITE` operations for `8086` microprocessor. 10

   b) Narrate the function of using `1`, `2` and `9` under `INT 21h` instruction. 6

   c) Distinguish between the followings: 9

      i. Polling and Interrupt.
      ii. Memory-mapped I/O and Isolated I/O.

302

5. a) Find out the similarity between the register sets of 8085 and 8086 microprocessors. 10
   b) Briefly explain the operations of IOPL and NT flags of 80286 microprocessor. 7
   c) To perform `MUL` and `DIV` operation, write two assembly language programs each for `MUL` and `DIV` using: 8
      i. 8086 Data Register Sets
      ii. 8086 Bit Manipulation Instructions

6. a) With an appropriate timing diagram clearly define the following terms: *Clock cycle*, *Machine cycle* and *Instruction cycle*. 9
   b) Differentiate between different `80x86` microprocessors. 9
   c) Derive the contents of the `IN AL, FFh` using the instruction template and also show how the contents of this instruction can be stored in memory. 7

7. a) What is Memory Segment? How is the main memory of 8086 processor segmented? 8
   b) Briefly explain the operations of a Program Counter. 8
   c) Write appropriate assembly language codes to accomplish the following tasks (use as many as possible arithmetic instructions with less number of registers): 9
      i. (30 + 15) * (575 – 225) + 210
      ii. 0Bh * (200 - 225) + 127
      iii. FFFh * 10h + 1111b

8. a) What are real mode, protected mode and virtual mode? Which microprocessor(s) first implements the virtual mode and how? 10
   b) Distinguish between the DX and SX version of 80386 microprocessor. 8
   c) Write an assembly language program structure to clearly state the operational differentiation between `LABEL` and `LOOP`? 7

B.Sc. Engg. / HD CSE 5th Semester (65) May 16, 2017

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2016-2017**
DURATION: 3 Hours **FULL MARKS: 150**

CSE 4503: Microprocessors and Assembly Language

Programmable calculators are not allowed. Do not write anything on the question paper.
There are **8 (eight)** questions. Answer any **6 (six)** of them.
Figures in the right margin indicate marks.

---

1. a) i. What is PLT and GOT? 6
   ii. What's the motivation for using PLT stub as function trampoline?
   iii. Why NOT call the shared library functions directly?
   b) Explain how PLT and GOT is used to resolve the base address for any function from any shared libraries which is compiled with Position Independent Code parameter. (3 states) 6
   c) What is meant by the term Buffer Overflow? How has this problem been resolved in the existing Linux library? 5
   d) Code to call a function with 3 parameters from "_start" stub in intel x86 architecture. Demonstrate the same code in x86_64 architecture. 8

2. a) What is Address Space Layout Randomization (ASLR)? Why Global Offset Table has the term 'Global' and 'Offset' in it? 4
   b) Show the byte representation of the following commands- 6
      i. MOV AX, [6072H]
      ii. MOV CX, DX
   c) Write two assembly programs, 15
   Program-I will take input and display the inputs (see test case) and store it in the stack. Consider the content of the stack won't change.
   Now, Program-II will take the values from stack and display the outputs as shown below.

Program-I		**Program-II**	
Enter Dividend:	10	Quotient:	3
Enter Divisor:	3	Remainder:	1

   Note: Dividend will be maximum of 2 digits and divisor of at maximum 1 digit.

3. a) What are NULL Bytes? Write down its significance in the buffer exploits? 6
   b) Explain steps of how interrupt is processed in 8086 microprocessors? How does 8086 get the address of any particular Interrupt Service Routine? 6
   c) How is the 'Interrupt Vector Table' and 'Interrupt Service Routine' related? Specify, what they hold and why. 6
   d) Explain Call and return mechanism for a near procedure and show how the IP and Stack are affected by procedure calls. 7

4. a) Analyze the given code: 4

```
section .data
1.      name db "/bin/sh"
section .text
2.      global _start

_start:
        push 0
```

3. push name
4. mov rax,59
5. mov rdi,rsp
6. mov rsi,0
7. mov rdx,0
8. syscall

Write comments for instructions 1 to 8. Mention for what purpose the instructions have been used. 6

b) Write down the significance of Accumulator register before and after function calls by conventional programming. 15

c) Remove the NULL bytes from the given stub to generate a new code in 32bit assembly which can be used as a shellcode to exploit buffer. [You can work with decimal of the respective hex values, i.e. 10d= 0xA]

shell: file format elf32-i386

Disassembly of section .text:

```
08048060 <_start>:
 8048060:   b8 66 00 00 00     mov   eax,0x66
 8048065:   bb 01 00 00 00     mov   ebx,0x1
 804806a:   6a 00              push  0x0
 804806c:   6a 01              push  0x1
 804806e:   6a 02              push  0x2
 8048070:   89 e1              mov   ecx,esp
 8048072:   cd 80              int   0x80
 8048074:   89 c2              mov   edx,eax
 8048076:   b8 66 00 00 00     mov   eax,0x66
 804807b:   bb 0e 00 00 00     mov   ebx,0xe
 8048080:   6a 04              push  0x4
 8048082:   54                 push  esp
 8048083:   6a 02              push  0x2
 8048085:   6a 01              push  0x1
 8048087:   52                 push  edx
 8048088:   89 e1              mov   ecx,esp
 804808a:   cd 80              int   0x80
 804808c:   b8 66 00 00 00     mov   eax,0x66
 8048091:   bb 02 00 00 00     mov   ebx,0x2
 8048096:   6a 00              push  0x0
 8048098:   66 68 2b 67        pushw 0x672b
 804809c:   66 6a 02           pushw 0x2
 804809f:   89 e1              mov   ecx,esp
 80480a1:   6a 10              push  0x10
 80480a3:   51                 push  ecx
 80480a4:   52                 push  edx
 80480a5:   89 e1              mov   ecx,esp
 80480a7:   cd 80              int   0x80
```

5. a) Draw and add a brief explanation of the flag register in 80286. What are the new flags introduced in 80386 and 80486? 6+4

b) Draw the flow diagrams for the three I/O data transfer techniques and label the states. 15

6. a) State the differences of Memory mapped I/O and Isolated I/O with figures. 4

b) "Software interrupts are prioritized more than the hardware interrupts when they occur at the same time" - justify the statement. 4

c) Suppose, a multicore processor was carrying out multiple instructions which is obvious. 9

Also, each of the core was responsible for parallel execution of multiple instructions. How was this possible?

Contrast the ideas of Pipelining and Superscalar property in microprocessors.

d) Briefly mention and explain the pipelining stages of Floating Point Unit. 8

7. a) What is an addressing mode? What are different addressing modes? Explain Based Indexed addressing modes. 8

b) Write some instructions to replace each upper-case letter in the following string by lower case equivalent using Base addressing mode. 7

MSG DB 'LIFE IS GOOD',$

c) The algorithm given below multiplies two unsigned numbers A and B. Using the algorithm, write a procedure MULTIPLY to multiply two numbers A and B. 10

[Assume, A and B are already stored in AX and BX.]

```
Product = 0
REPEAT
        IF LSB of B is 1
        THEN
        Product = Product + A
        END_IF
        Shift left A
        Shift right B
UNTIL B=0
```

8. a) Given, 7

MOV EAX, 0x2002b2(RIP)

Accessing the offset from RIP register is possible only in **x86_64** architecture, show how this is handled in **x86** architecture.

b) What is the necessity of using stack segment in assembly language programming? With suitable examples demonstrate how to perform push and pop operation in stack. 6

c) What's the purpose of stack pointer(SP) and Base Pointer (BP) registers? Show the changes in the stack as the function foo executes. Show each state of changes till the stack becomes empty again. Initial stack state is NULL. 2+10

```
int main() {
        foo(3,'A',10);
        return 0;
}
```

```
void foo(int a, char b, int c){
    int k;
    k= a;
    if(k>0){
        k--;
        foo(k,'A',10);
    }
    else return;
}
```